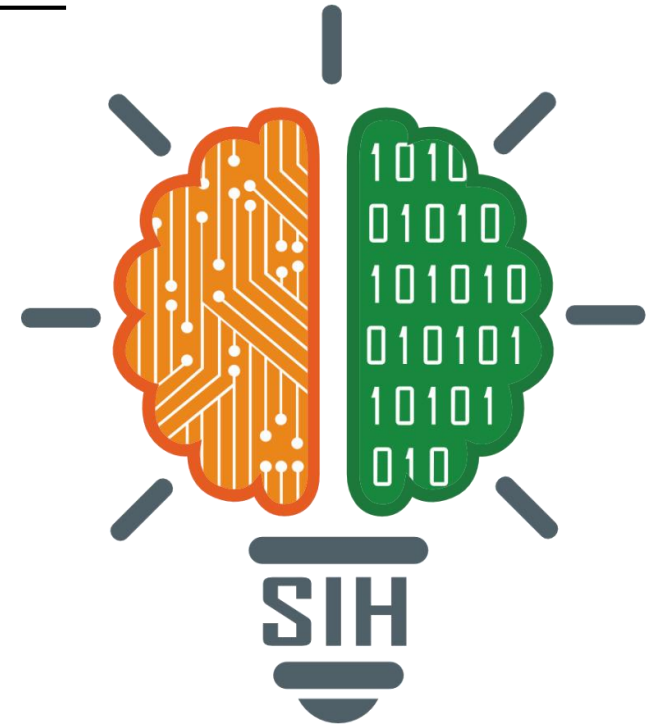


TideTrail SRM: Deep Learning Super-Resolution of Sentinel-2 Imagery, 10 m to 2.5 m

- Problem Statement ID - SIH26142
- Problem Statement Title- Deep Learning Based Super Resolution Mapping (SRM) from Medium Resolution Satellite Imageries
- Theme- Space Technology
- PS Category- Software
- Team ID- SIH361
- Team Name - Meridian



TideTrail SRM: Super-Resolution Mapping of Sentinel-2 Imagery

• How It Addresses the NTRO Problem Statement

- **4x Sharper, Spectrally Faithful:** Sentinel-2 blue, green, red and NIR go from 10 m to 2.5 m on an exact grid. Averaged back to 10 m, the result matches the satellite pixel (error 0.0001).
- **Validated, Not Claimed:** 4 held-out scenes in CO, LA, MN, scored against 0.6 m aerial imagery: 37.25 dB PSNR, bicubic 37.00 dB.
- **Error Is Accounted For:** every pixel carries an uncertainty value; it ranks real error (rank correlation 0.24) and its 90% intervals held 91% of the time.

• Detailed Explanation of the Proposed Solution

Step 1 Data In: Sentinel-2 L2A bands B2, B3, B4 and B8 at 10 m, uploaded as a GeoTIFF.

Step 2 Pre-process: reflectance, cloud screening, sub-pixel registration and radiometric matching to aerial references.

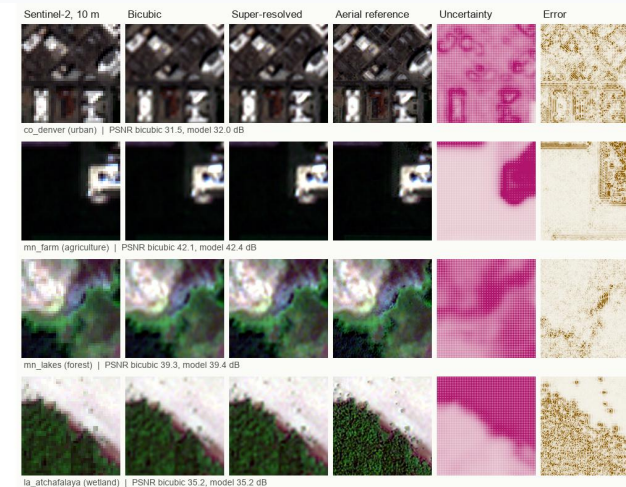
Step 3 Super-resolve: residual CNN with sub-pixel convolution, tiled inference, uncertainty head.

Step 4 Validate: PSNR, SSIM, spectral angle, NDVI, water and edge checks on regions the model never saw.

• Innovation and Uniqueness of the Solution

- **Detail On A Safe Base:** the network adds learned detail to a bicubic enlargement, so it starts from a faithful image instead of inventing one.
- **Consistency Step:** after inference every 10 m pixel is moved back to what the satellite measured; only the detail inside it is kept.
- **Doubt Is Shown:** a second map marks where detail is inferred, not observed, so analysts can discount it.

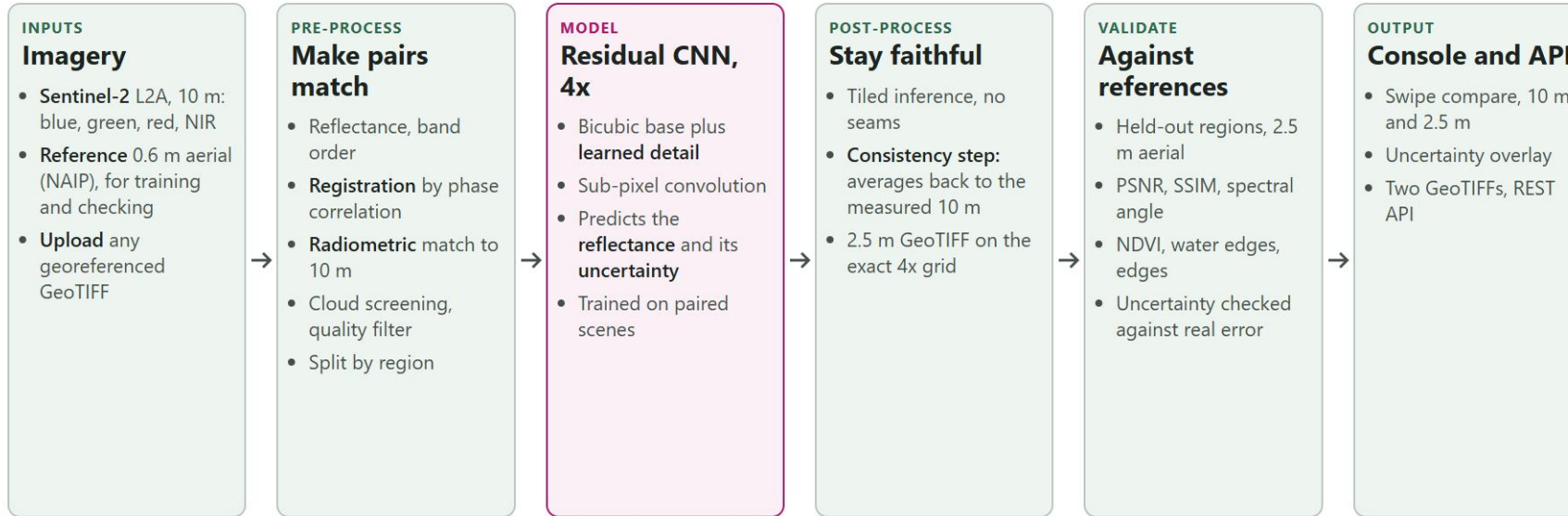
Prototype: 10 m and 2.5 m, Swipe to Compare



TECHNICAL APPROACH (TECHNOLOGIES, PROCESS & WORKING PROTOTYPE)



• Architecture Pipeline



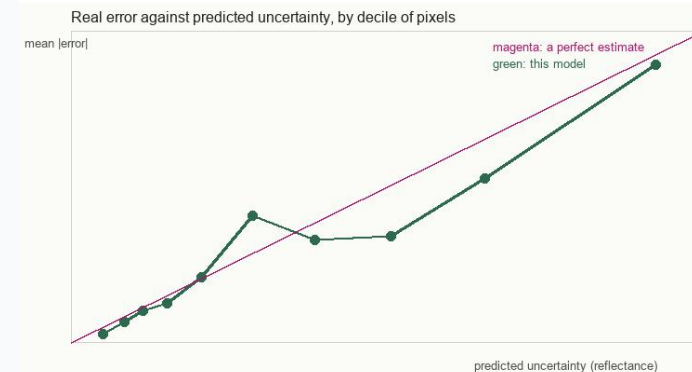
Runs offline on a laptop CPU, no GPU

10 m to 2.5 m, 16 times the pixels

• Technologies to be Used

- **Deep Learning:** PyTorch residual CNN, 0.69 M parameters, sub-pixel convolution, Laplace uncertainty head; trained on 27 Sentinel-2 and aerial pairs.
- **Geospatial & Backend:** rasterio on Cloud Optimized GeoTIFFs, phase-correlation registration, FastAPI; runs fully offline on a CPU.
- **Visual Console:** Vanilla HTML5/ES6, Leaflet 1.9, swipe compare, NDVI and water layers, uncertainty overlay.

• Prototype: Uncertainty and Accuracy Panels





1. Compute & Speed	• under 45 s to enhance a 3.2 km scene on a laptop CPU; no GPU needed. • Training is resumable on the same CPU; this checkpoint is 250 iterations.
2. Data Access	• Free and open: Sentinel-2 L2A from Copernicus, NAIP aerial references from USDA. • NTRO's own high-resolution imagery can serve as the reference.
3. Deployment	• REST API and GeoTIFFs on the input's own grid, for existing GIS tools. • Standalone Reliability: Runs completely on-premise without cloud latency.

• Challenge 1: Invented Detail

The Risk: a sharper image can contain texture the satellite never saw, and mislead an analyst.

Our Solution: learned detail on a bicubic base, a step that keeps every 10 m value, and an uncertainty map that marks inferred detail.

• Challenge 2: The Reference Disagrees

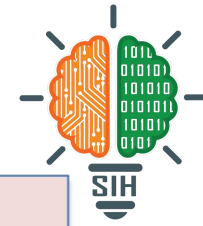
The Risk: aerial and satellite images differ in date, sun angle and registration; averaged to 10 m the aerial image sits 0.05 reflectance from Sentinel-2.

Our Solution: sub-pixel registration, per-band radiometric matching, cloud screening, and a reference anchored to the Sentinel-2 pixels.

• Challenge 3: Generalising to India

The Risk: free aerial references exist only in the US, so Indian scenes cannot be scored directly.

Our Solution: region-held-out testing in states the model never saw, an uncertainty map that reports doubt, and retraining on NTRO references when supplied.

**Operational Impact:****NTRO Intelligence**

- Sharper 2.5 m views from free 10 m imagery, doubt marked.

Crop Monitoring

- Field-scale NDVI; error against the aerial reference 0.026, bicubic 0.027.

Urban Analysis

- Edge F1 0.45 against 0.41 for bicubic: buildings and roads are crisper.

Disaster Assessment (NDMA)

- Water-edge overlap 0.84 against 0.81 for bicubic, for flood and shoreline mapping.

• Environmental Value

- **Crop Stress:** Spots field level change early.
- **Water Extent:** Tracks floods and shrinking wetlands.

• Economic & Resource ROI

- **No Imagery Fees:** Works on free Sentinel-2 data.
- **Better Triage:** Aims costly high-resolution tasking.

• Social & Planning Value

- **Disaster Response:** Faster, sharper flood and damage maps.
- **Local Planning:** Street-level views of growing towns.

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